

- (v) As m_2 groot is, sal m_1 neig om op te beweeg.
 With m_2 large, m_1 would tend to move upwards.

Die som van die kragte in die x-rigting op m_1 (gee in simbole): / The sum of the forces in the x-direction on m_1 (give in symbols):

$$+ T_{m_2} - F_s - m_1 g \sin \theta = 0 \quad \checkmark$$

- (vi) Bepaal nou m_2 met $m_1 = 10$ kg. / Now, determine the value of m_2 with $m_1 = 10$ kg

$$m_2 g - (F_s - m_1 g \sin \theta) = 0$$

$$m_2 (9.8) - (0.4 \cdot 9.8 \cdot 10 \cos 37) - 10 \cdot (9.8) \sin 37 = 0$$

$$m_2 (9.8) = 90.284$$

$$m_2 = 9.2 \text{ kg} \quad \checkmark$$

- (vii) As $m_2 = m_1 = 10$ kg. en $\mu_k = 0.3$ is, toon aan die versnelling van die sisteem is $a = 0.78 \text{ m/s}^2$.
 With $m_2 = m_1 = 10$ kg. and $\mu_k = 0.3$, show that the acceleration of the system is $a = 0.78 \text{ m/s}^2$.

$$+ T_{m_2} - F_s - m_1 g \sin \theta = m_2 a$$

$$10 \cdot 9.8 - (0.4 \cdot 9.8 \cdot 10 \cos 37) - (10) \cdot 9.8 \sin 37 = 10 \cdot a$$

$$98 - 90.2843 = 10a$$

$$a = 0.78 \text{ m/s}^2 \quad \checkmark$$

(6)

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